



**Parvaj Malik**  
Bachelor of Technology  
Electronics and communication  
HNBGU, Central University

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## EDUCATION

| Degree/Certificate | Institute/Board                                                   | CGPA/Percentage | Year     |
|--------------------|-------------------------------------------------------------------|-----------------|----------|
| B.Tech, ECE        | Hemwati Nandana Bahugunua Garhwal Central university, Uttarakhand | 7.1             | May 2024 |
| Senior Secondary   | State Board                                                       | 81.2            | Apr 2019 |
| Intermediate       | State Board                                                       | 80              | Apr 2019 |

## EXPERIENCE

- Inochi Care Private Limited – Medical Device Division**   
Feb 2025 – Present  
Delhi, India
  - Embedded Systems Engineer – R&D (July 2025 – Present)**
    - Transitioned from IIT Delhi–AIIMS collaborative biomedical research project into product development role.
    - Leading firmware development for Negative Pressure Wound Therapy (NPWT) device.
    - Architected STM32/NXP-based real-time pressure control system using RTOS.
    - Developed LVGL-based GUI and contributed to PCB validation and optimization.
    - Built integrated hardware-software testing jigs for automated validation and calibration.
    - Collaborating with AIIMS clinical partners for device testing and deployment.
  - Research Embedded Systems Engineer – IIT Delhi**  (WIPO-Supported Biomedical Project) (Feb-July)
    - Developed EMG and Tocography signal acquisition firmware under WIPO-supported research.
    - Designed analog front-end and multi-channel data acquisition architecture.
    - Contributed to clinical validation and biomedical system prototyping.
    - Contributed to development of a gait analysis system using multi-channel pressure sensors for clinical diagnostics.
    - Engineered a wearable nasal respiration monitoring device from scratch, including sensor integration and real-time data acquisition

## CDAC

Aug 2024 – Feb 2025  
Pune, India

- Embedded Design Systems Training**
  - Trained in Embedded C, C++, Python, and Data Structures.
  - Hands-on development with STM32F407, ESP32, ESP8266, Vega processor, and Raspberry Pi.
  - Worked extensively with UART, SPI, I2C, CAN, ADC, and Timers.
  - Implemented Linux IPC, sockets, kernel compilation, pseudo drivers, and RTOS on STM32.
  - Developed IoT systems integrating sensors, Wi-Fi, BLE, and LoRaWAN.

## PROJECTS

- Project 1: Negative Pressure Wound Therapy Device: Ongoing Project**  
*Tools: STM32CubeIDE, MCUExpresso, LVGL (GUI Development), RTOS, KICAD* 
  - NPWT device uses smart pressure and sensor technology to promote faster wound healing.
  - Designs and optimizes firmware that bridges hardware and software to deliver precise, real-time performance.
  - Developed a custom PCB for compact and wearable integration of the device, ensuring reliability and comfort.
- Project 2: Nasal Monitoring System: [A wearable device]**  
*Tools: ESP32, BME280, Arduino IDE, MQTT, Cloud Storage, Web server* 

- Developed a wireless nasal respiration monitor to track inhale and exhale patterns.
  - Implemented MQTT protocol to send sensor data to a cloud server for remote monitoring. firmware using ESP32 for real-time sensor data acquisition
  - Developed a custom PCB for compact, wearable integration of the device.
- **Project 3: Electromyography (EMG) Detection Device [A wearable Belt]**  
*Tools: ESP32, AD620, ADS1115 ADC, Arduino IDE, Plotter, BLE, MATLAB, PCB Software* 
  - Developed a system to capture and analyze muscle signals (EMG) for physical therapy and rehabilitation applications
  - Using the AD620 instrumentation amplifier and implemented digital filtering for noise reduction.
  - ADS1115 ADC for high-resolution analog-to-digital conversion of EMG signals.
  - Used ESP32 to wirelessly transmit EMG data via Bluetooth Low Energy (BLE).
  - A real-time data interface using MATLAB for graph visualization and signal analysis.
- **Project 4: Landmine Detection Robotic Vehicle**  
*Tools: STM32, GPS, GSM, Ultrasonic Sensor, Metal Detector, L298N* 
  - Developed a landmine-detecting robot with remote alert functionality.
  - Integrated metal detector and ultrasonic sensors for obstacle and mine detection. GPS + GSM communication system to transmit real-time coordinates.
  - Applied PWM motor control logic for smooth directional control
- **Project 5: Smart Light Control System**  
*Tools: STM32, LDR Sensor, Relay Module, LCD, Circuit designing, Switch button* 
  - STM32 microcontroller to control system logic and display real-time status on an LCD.
  - Implemented threshold-based decision logic to switch lights on/off automatically. threshold analysis to determine switching points -HAL ADC functions to measure the analog value representing ambient light intensity.
- **Project 6: Gait Analysis System (Ongoing) [Analyze walking patterns through foot pressure sensors.]**  
*Tools: Arduino, Force Sensors(FSR), UART, Multiplexor, Design* 
  - Real-time foot pressure detection using microcontroller-based sensors
  - Implemented data transmission and visualization for clinical gait diagnostics. Created firmware to process and transmit multi-channel sensor data.
  - Currently deployed and actively used by doctors at AIIMS Delhi for gait diagnostics and rehabilitation tracking
- **Project 7: Obstacle Avoiding Robot [Robotic car]**  
*Tools: Arduino, Ultrasonic Sensor, Bluetooth Module, Hardware part, Matlab* 
  - Real-time foot pressure detection using microcontroller-based sensors
  - Implemented data transmission and visualization for clinical gait diagnostics. Created firmware to process and transmit multi-channel sensor data.
  - Currently deployed and actively used by doctors at AIIMS Delhi for gait diagnostics and rehabilitation tracking
- **Project 8 : Multi-LED Control with Switch Interrupt**  
*Tools: STM32, Push Button, LED's, Web Server, Wi-Fi module* 
  - Interrupt-driven LED control system using STM32 and a single push button.
  - Integrated a Wi-Fi module to transmit LED status data to a web server in real times. Designed a simple web interface to visualize LED states remotely
  - Applied interrupt-based design to reduce CPU load and improve system responsiveness.

SKILLS

- **Programming Languages:** [Embedded C, CPP, Python, Product Firmware]
- **Technologies:** [Micro-controllers(STM32, NXP MIMXRT1160,66,ESP32, ESP8266, Vega processor, Beagle bone), UART, I2C, SPI, CAN, Operating system, RTOS with STM32, PCB Designing, Device driver, Kernel compilation, IOT, BLE, LoraWan, Circuit Designing, Digital]
- **Tools:** [VScode, STM32CubeIDE, MCUExpresso IDE, Arduino IDE, MATLAB, Ki-CAD]

CERTIFICATIONS

- **CDAC Training**, [Certification 1](#) Aug 2025
- **State Level Scout guide**, [Certification 2](#) 2016

POSITIONS OF RESPONSIBILITY

- **Position A**, Placement Cell coordinator, HNBGU 2024
- **Position B**, Founder and Lead Organizer Technical Fest (2023, 2024)